

Targeted therapies and immunotherapy for acute lymphoblastic leukaemia

Targeted and immunotherapy drugs may be used to treat ALL. These drugs target something in or around the leukaemia cell that is helping it to grow or use the immune system to attack cancer cells.

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About targeted therapies and immunotherapy for acute lymphoblastic leukaemia (ALL)

You may have a targeted therapy or immunotherapy drug, depending on the [type of ALL](#) you have. These drugs target something in or around the leukaemia cell that is helping it to grow. Some targeted drugs are also immunotherapy drugs. They use the immune system to find and attack cancer cells.

Your doctor or nurse will explain which type of drug is suitable for you.

Some of these drugs may only be available in certain situations. If a drug is not available to you on the NHS, there may be different ways you can still have it. Your cancer doctor can give you advice. We have more information about what to do [if a treatment is not available](#).

Tyrosine kinase inhibitors (TKIs)

TKIs are a type of targeted therapy drug used to treat Ph+ ALL. Tyrosine kinase (TK) is a chemical messenger that tells the leukaemia cells when to develop and divide. In [Ph+ ALL](#), the leukaemia cells make high levels of tyrosine kinase. This makes them grow in an uncontrolled way. TKI drugs block (inhibit) the TK signal, which causes the leukaemia cells to die.

You have a TKI with [chemotherapy](#) as your first treatment. They are also used if the leukaemia does not respond to treatment or comes back. You take them as tablets or capsules every day.

Types of TKIs

- [imatinib](#)

- [dasatinib](#)
- [ponatinib](#)
- [nilotinib](#).

Imatinib is the most often used TKI. You usually have it first.

Side effects of TKIs

Side effects of TKIs are usually mild. They include:

- feeling sick
- diarrhoea
- tiredness
- leg aches or cramps
- an itchy rash.

These are usually more noticeable in the first 4 weeks of treatment. After that, they begin to get better. Tell your nurse or doctor if you have any side effects. There is usually something they can do to treat them.

Monoclonal antibodies

Monoclonal antibodies are targeted therapy drugs that work on the immune system. They target certain proteins on the leukaemia cells and use the immune system to attack them.

You have these drugs with chemotherapy or on their own. You have them as a drip into a vein.

[Rituximab](#) is used to treat [B cell ALL](#) if the leukaemia cells have a protein called CD20 on their surface. This is sometimes called CD20-positive leukaemia. You may be able to have rituximab as an injection under the skin after you have had some treatments as a drip.

Other monoclonal antibodies may also be used to treat B cell ALL. They target different proteins found on the leukaemia cells:

- [Blinatumomab](#) (Blinicyto[®]) may be used if the leukaemia cells are CD19-positive. You may have it if you are in [remission](#) but have very small numbers of leukaemia cells (minimal residual disease).
- Inotuzumab ozogamicin (Besponsa[®]) may be used if the leukaemia cells are CD22-positive. You may have it if ALL does not go into remission with chemotherapy or if it has come back (relapsed).

Your doctor or specialist nurse can tell you more about these drugs and their side effects.

CAR-T therapy

CAR-T therapy (or [CAR T-cell therapy](#)) is an immunotherapy treatment. CART T stands for chimeric antigen receptor T-cell. This treatment involves collecting your own immune cells and changing them so they become better at fighting the leukaemia cells. These cells then target the leukaemia cells.

The type of CAR-T therapy used to treat ALL is called tisagenlecleucel (Kymriah[®]). It may be used to treat young people under 25 with B cell ALL if other treatment has not worked, or if ALL comes back.

CAR-T therapy is complicated and can have serious side effects. It is only suitable for a very small number of people. Currently, it is only available in a few specialist hospitals. Some people may be offered it as part of a [clinical trial](#).

About our information

References



Below is a sample of the sources used in our acute lymphoblastic leukaemia (ALL) information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

NICE (National Institute for Health and Care Excellence). Blood and bone marrow cancers. Available from <https://pathways.nice.org.uk/pathways/blood-and-bone-marrow-cancers> [accessed August 2021].

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help